

AI Prompt Log Template

Team Name: [Your Team Name]

Project: [Brief project title]

Instructions: Document all significant AI interactions throughout your project. This log demonstrates transparency and helps you reflect on the AI's usefulness.

Log Entry Template

Copy this template for each major AI interaction:

Entry [Number]

Date: [MM/DD/YYYY]

AI Tool: [ChatGPT / Claude / Gemini / Copilot / Other]

Task/Goal: [What were you trying to accomplish?]

Prompt Used:

[Paste the exact prompt you used]

AI Response Summary: [Briefly summarize what the AI provided - you can paste key portions or screenshots]

What You Trusted: [What parts of the AI's response did you use directly?]

What You Changed/Corrected: [What did you modify and why?]

Verification Method: [How did you verify the AI's output? Cross-checked with textbook? Tested code? Compared with other AI?]

Usefulness Rating: [1-5 stars] ★★★★★

Notes: [Any other observations about this interaction]

Example Entry

Entry 1

Date: 10/15/2025

AI Tool: ChatGPT

Task/Goal: Understand assumptions for Pearson correlation

Prompt Used:

```
I'm analyzing the relationship between screen time (hours/day) and sleep
quality (1-10 scale)
using Pearson correlation. What assumptions do I need to check before running
this analysis?
Please explain each assumption and how to test it in R.
```

AI Response Summary: ChatGPT listed four assumptions: (1) continuous variables, (2) linear relationship, (3) no outliers, (4) bivariate normality. It provided R code for checking linearity with scatterplots and testing normality with Shapiro-Wilk test.

What You Trusted: The list of assumptions was accurate and matched our textbook. The scatterplot code worked perfectly.

What You Changed/Corrected: The AI suggested Shapiro-Wilk for normality, but our textbook also mentioned Q-Q plots as more visual, so we added that. Also, the AI initially said “homoscedasticity” which isn’t required for correlation (that’s for regression), so we clarified this with Claude.

Verification Method: Cross-checked assumptions against textbook Chapter 9 and course lecture slides. Ran the suggested R code on our data to confirm it worked.

Usefulness Rating: ★★★★★ (4/5)

Notes: ChatGPT was very helpful but sometimes mixes up regression and correlation assumptions. Good to verify with multiple sources.

Tips for Effective Prompt Logging

1. **Log as you go** - Don't wait until the end to reconstruct your prompts
 2. **Be specific in prompts** - Better prompts get better responses
 3. **Compare AI tools** - Try the same prompt with different AIs to compare responses
 4. **Document failures too** - Show what didn't work, not just successes
 5. **Note your learning** - What did you learn from each interaction?
 6. **Save screenshots** - For complex outputs (graphs, tables), screenshots are helpful
 7. **Track time** - Note if an AI saved time or caused confusion
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Section-by-Section Log Organization

You may want to organize your log by project phase:

Phase 1: Understanding the Method

[Log entries about learning the statistical concept]

Phase 2: Data Preparation

[Log entries about data cleaning, exploration, assumption checking]

Phase 3: Analysis Execution

[Log entries about running the analysis, interpreting output]

Phase 4: Visualization and Presentation

[Log entries about creating figures and preparing presentation]

Summary Comparison Table

At the end of your log, include a summary comparing the AI tools:

Aspect	ChatGPT	Claude	Gemini	Copilot
Used for what?				
Strengths				
Weaknesses				
Best at				
Worst at				
Overall rating	★★★★★	★★★★★	★★★★★	★★★★★

Reflection Questions

After completing your project, reflect on:

1. Which AI was most helpful overall and why?
2. What task was AI best at helping with?
3. What task was AI worst at or caused confusion?
4. How did you develop better prompts over time?
5. What would you do differently next time?
6. What’s one specific recommendation for using AI in statistics?

Submission Checklist

- ☐ All major prompts documented with dates
- ☐ At least 2 different AI tools compared
- ☐ Verification methods described
- ☐ Both successes and failures included
- ☐ Summary comparison table completed
- ☐ Reflection questions answered
- ☐ Screenshots attached (if applicable)

Total Log Entries: [Number]

Primary AI Tools Used: [List]

Submission Date: [Date]